Matrix Theory

09 Linear Independence, Basis, and Dimension

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From the last class

- Elimination reduces $Ax = b$ to $Ux = c$ and $Rx = d$

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- $m - r$ solvability constraints on $b/c/d$

$A_{m \times n}$

r

$m - r \left\{ \begin{bmatrix} \cdots \\ 0 & 0 & 0 \end{bmatrix} \right.$ $\quad 0$

From the last class

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- The rank of those matrices (A, U, R) is r

Going forward

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- Gave r pivot rows and r pivot columns
- Null space is combination of $n - r$ special solutions
- $m - r$ solvability constraints on $b/c/d$
- The rank of those matrices (A, U, R) is r

Rank r is crucial!

True size of the system

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

- m and n do not give the complete picture

True size of the system

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

$R_3 = 2R_2 - 5R_1$

$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

- m and n do not give the complete picture
- We have seen R_3 is a combination of R_1 and R_2 (elimination led to zero row)

True size of the system

$$A = \begin{array}{c} \\ \\ \\ \end{array}\begin{array}{cccc} C_1 & & & C_4 \\ \end{array}$$

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

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- Columns also failed to be independent; $C(A)$ became $2D$ plane

True size of the system

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \longrightarrow 2$$

- m and n do not give the complete picture
- We have seen R_3 is a combination of R_1 and R_2 (elimination led to zero row)
- Columns also failed to be independent; $C(A)$ became $2D$ plane
- Rank (true size) emerges to be an important number!

Rank: computational definition

$A \xrightarrow{E} U$

- Number of pivots in the elimination process

Rank: computational definition

- Number of pivots in the elimination process
- Number of nonzero rows in U

Rank: computational definition

- Number of pivots in the elimination process
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Rank counts the number of independent rows in the matrix!

Linear Independence

Given a set of vectors, $v_1, v_2, \ldots, v_n$, and if their combination $c_1v_1 + c_2v_2 + \ldots + c_nv_n = 0$ only when $c_1 = c_2 = \ldots = c_n = 0$, then the vectors $v_1, v_2, \ldots, v_n$ are linearly independent.

Linear Independence

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And if any $c_i \neq 0$, the vectors are dependent. One vector is a combination of others!

Linear Independence

A collection is linearly independent when every vector contributes something that cannot be produced by the others.

Example-1

- If any $v_i = \bar{0}$, the set is linearly dependent

$\{v_1 \quad v_2 \quad \dots \quad v_n\}$

$\underline{3} \cdot \bar{0} + \underline{0} \cdot v_2 + \dots + \underline{0} \cdot v_n = 0$

dependent

Example-2

- Columns $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Columns $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0$$

$$c_1 + c_2 = 0 \qquad c_1 = 0$$

$$c_2 = 0$$

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 0$$

$$c_1 + 3c_2 = 0 \Rightarrow c_1 = -3c_2$$

$$2c_1 + 6c_2 = 0$$

$$(-3, 1) \qquad (-6, 2)$$

Example-3

- Columns $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

$V_1 + V_2 = V_3$

Combination $\iff$ Dependent

$$\sum_{i=1}^{n} c_i v_i = 0$$

not all $c_i = 0$

A set of vectors is dependent iff at least one vector can be written as a combination of the others.

$$v_i = \sum_{j=1 (\neq i)}^{n} c_j v_j$$

$$-1 \cdot v_i + \sum_{j=1 (\neq i)}^{n} c_j v_j = 0$$

Independence and Null space

- Put the vectors into the columns of a matrix $\begin{bmatrix} \vdots & \vdots & & \vdots \\ v_1 & v_2 & \dots & v_n \\ \vdots & \vdots & & \vdots \end{bmatrix}$

Independence and Null space

- Put the vectors into the columns of a matrix $\begin{bmatrix} \vdots & \vdots & & \vdots \\ v_1 & v_2 & \dots & v_n \\ \vdots & \vdots & & \vdots \end{bmatrix}_{m \times n}$
- $Ax = x_1v_1 + x_2v_2 + \ldots + x_nv_n = \bar{0}$

$N(A)$

$\{x \in \mathbb{R}^n : Ax = 0\}$

0

$\{\bar{0}\}$

Independence and Null space

- Put the vectors into the columns of a matrix $\begin{bmatrix} \vdots & \vdots & & \vdots \\ v_1 & v_2 & \dots & v_n \\ \vdots & \vdots & & \vdots \end{bmatrix}$
- $Ax = x_1v_1 + x_2v_2 + \ldots + x_nv_n$
- Vectors are independent $\iff$ $Ax = 0$ has only $x = 0$

Echelon Matrix

$$\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

- r nonzero rows are independent

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} = 0$$

$c_2 = 0$

$c_1 = 0$

$c_1 (1, -2)$

$c_2 (0, 1)$

$c_1 \quad -2c_1 + c_2 = 0$

$c_1 \Downarrow 0$

$c_2 = 0$

Echelon Matrix

- r nonzero rows are independent
- r columns that contain the pivots are independent

Dependent or Independent?

$A_{m\times n}$ $\quad \mathbb{R}^2$

$m<n$

- A set containing 0? D
- One nonzero vector? I
- Two identical vectors? D
- Two nonzero scalar multiples? D
- More than n vectors in $\in \mathbb{R}^n$

$$A \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

$$E \downarrow$$

$$U \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$n \times m \qquad m \le n$

$2c_1 + c_2 = c_3$

[Short & wide] $n < m$ $\quad (m-n)$

$Ax = 0$ nontrivial

Triangular Matrix

$$\begin{bmatrix} 3 & 4 & 2 \\ \cancel{0} & 1 & 5 \\ \cancel{0} & \cancel{0} & 3 \end{bmatrix}$$

$v_1 \quad v_2 \quad v_3$

U

$$c_1 \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix} = 0$$

$c_3 = 0$

$c_2 + 5c_3 = 0 \Rightarrow c_2 = 0$

$\Rightarrow c_1 = 0$

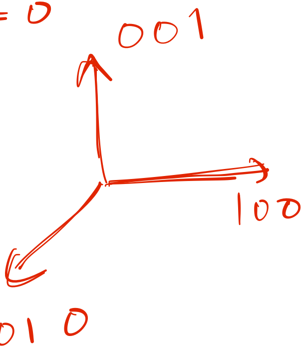
Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

I_4

$c_1 = c_2 = c_3 = c_4 = 0$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$



$= b_1 e_1 + b_2 e_2 + b_3 e_3$

When $m < n$

- Set of n vectors in $\mathbb{R}^m$ must be dependent

$A_{m \times n}$

$(n-r)$ special variables

$\Rightarrow$ non trivial solutions

When $m < n$

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- $A_{m \times n}$ is a fat matrix

When $m < n$

- Set of n vectors in $\mathbb{R}^m$ must be dependent
- $A_{m\times n}$ is a fat matrix
- $Ax = 0$ has nontrivial solutions

IND =)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Redundency

Spanning a space

If a vector space V consists of all linear combinations of $w_1, \ldots, w_l$, then these vectors span the space. Every vector v in V is some combination of the w's: $v = c_1 w_1 + \ldots + c_l w_l$ for some c_i's.

V :

$c_1 \quad \ldots \quad c_l$

$d_1 \quad \quad d_l$

$$\begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{matrix}$$

Spanning a space

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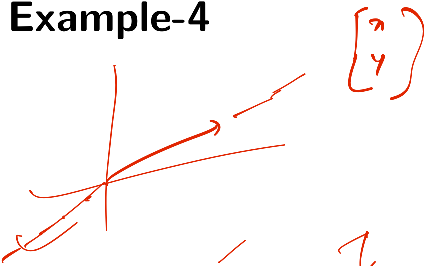
If a vector space V consists of all linear combinations of $w_1, \ldots, w_l$, then these vectors span the space. Every vector v in V is some combination of the w's: $v = c_1 w_1 + \ldots +_l w_l$ for some c_i's.

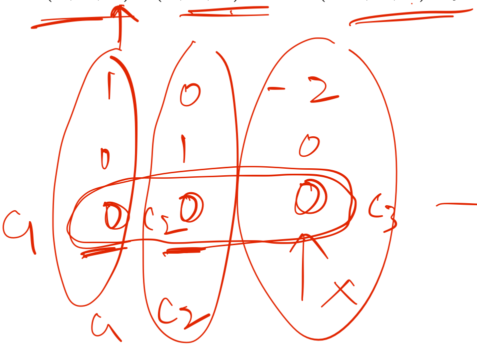
The c's need not be unique (different combinations are permitted; spanning set can be large!).

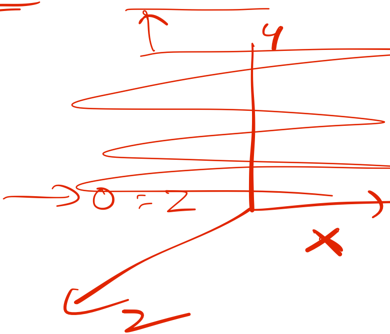
Example-4



$A \xrightarrow{E} U$

- Vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(-2, 0, 0)$ span the xy-plane in $\mathbb{R}^3$





Column space: span of columns

$$\begin{bmatrix} x_1 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

- $C(A) = \{Ax : \ \ x \in \mathbb{R}^n\}$

Column space: span of columns

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- $C(A) = \{Ax : \quad x \in \mathbb{R}^n\}$
- Space spanned by the columns of A

Independence + Span → Basis

- Consider $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$

Independence + Span → Basis

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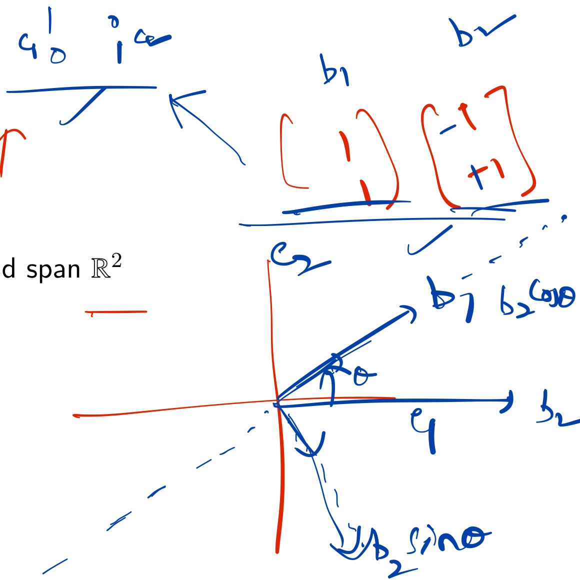
- Consider $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$
- They are independent and span $\mathbb{R}^2$



Basis

A set $\{v_1, v_2, \ldots, v_n\}$ is a basis for a vector space V, if it spans V and is linearly independent.

$\{ - \; - \; - \} \qquad \in \mathbb{R}^n$

Basis

A set $\{v_1, v_2, \ldots, v_n\}$ is a basis for a vector space V, if it spans V and is linearly independent.

Basis = enough vectors to generate everything, but no unnecessary vectors.

Basis

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A set $\{v_1, v_2, \ldots, v_n\}$ is a basis for a vector space V, if it spans V and is linearly independent.

Basis = enough vectors to generate everything, but no unnecessary vectors.

Every vector in the space is a combination of the basis vectors, because they span.

Basis expresses each vector uniquely

There is a unique way to write v as a combination of the basis vectors.

Multiple bases for the same space

- Consider $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

Multiple bases for the same space

- Consider $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$
- Both span $\mathbb{R}^2$. Which is correct?

Multiple bases for the same space

Bases

- Consider $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$
- Both span $\mathbb{R}^2$. Which is correct?
- Basis is not unique! A vector space has infinitely many different bases.

ind → removes redundancy

span → cover every thing

{Basis} → { . . . } { . . . }

What is common across different bases of a space?

The vectors in a basis are not intrinsic to the space. But the number of vectors in every basis is intrinsic.

Dimension of a space

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Any two bases for a vector space V contain the same number of vectors. This number, which is shared by all bases and expresses the number of "degrees of freedom" of the space, is the dimension of V.

Dimension is the same despite different bases!

If $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$ are both bases for the same vector space, then $m = n$. The number of vectors is the same.

$|\{v_1 \ldots v_m\}| = m$ $\quad |\{w_1 \ldots w_n\}| = n$

$m < n$

$w_1 = a_{11}v_1 + a_{21}v_2 + \cdots + a_{m1}v_m$

$w_n = a_{1n}v_1 + a_{2n}v_2 + \cdots + a_{mn}v_m$

$$\begin{bmatrix} | & | & & | \\ w_1 & w_2 & \cdots & w_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_m \\ | & | & & | \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & \vdots \\ \vdots & & & \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix}$$

$* \times 1$ $\quad * \times n$ $\quad * \times m$ $\quad m \times 1$

$\rightarrow$ Contradiction

$Ax = 0$

$VAx \quad Wx = 0$

$W = VA$

$* \times m \quad m \times n$

$VAx = 0$

Zero matrix is exceptional!

- Column space is $\{\bar{0}\}$

Basis ?

$\{\ \}$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$0_{3\times3}$

Zero matrix is exceptional!

- Column space is $\{\bar{0}\}$
- By convention, the basis is $\{\}$ and dimension is zero

Next

- The Four Fundamental Subspaces of a matrix!

Rough

$\quad 1 \qquad -1 \qquad -1$

1) $\{1+x, \; x+x^2, \; 1-x^2\}$: $P_2 = \{1, x, x^2\}$

2) $\sin^2 x, \; 1, \; \cos^2 x$

$\quad 1 \qquad -1 \qquad 1$

$a_0 + a_1 x + a_2 x^2$